

```
xs_squared = [x**2 for x in xs]
xs_threehalves = [x**1.5 for x in xs]
xs_cubed = [x**3 for x in xs]
plt.plot(xs,ys)
axes = plt.gca()
axes.set_ylim([np.min(ys),np.max(ys) + 140])
plt.plot(xs,xs_exp)
plt.plot(xs,xs)
plt.plot(xs,xs_squared)
plt.plot(xs,xs_cubed)
plt.plot(xs,xs_threehalves)
plt.title('Comparing Insertion Sort to Other Growth Rates')
plt.xlabel('Number of Files in Random Cabinet')
plt.ylabel('Steps Required to Sort Cabinet')
plt.show()
```

This results in [Figure 4-3](#).